SEASONAL FLU VACCINE PREDICTION

A Machine Learning Classification Project

Following the CRISP-DM Framework | Kenyan Public Health Context

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INTRODUCTION

Every flu season Kenya loses lives to a preventable disease. We built a machine learning model to identify who is least likely to vaccinate so that public health departments and hospitals can intervene before it is too late.

Using survey data from 26,707 respondents we identified the strongest predictors of vaccination behavior and generated specific recommendations for two stakeholders.

CRISP-DM FRAMEWORK

01 Business Understanding

Stakeholder, problem statement and why ML

02 Data Understanding

Dataset overview and key findings

03 Data Preparation

Cleaning, feature selection and preprocessing

04 Modeling

Five models built iteratively

05 Evaluation

Results and best model selection

06 Recommendations

Actionable insights for stakeholders

BUSINESS UNDERSTANDING

The Problem

Kenya has two flu seasons annually (March-May and October-December). Seasonal flu vaccination rates remain low particularly among younger and lower income populations.

Can we predict who will NOT get vaccinated so we can intervene proactively?

Stakeholder 1

County Director of Health

Identify WHICH communities to target for seasonal flu outreach every flu season

Stakeholder 2

CEOs; Kenyatta National Hospital & Mbagathi County Hospital

Identify WHICH patients to flag for additional clinical encouragement

DATA UNDERSTANDING

26,707
Survey Respondents

35
Original Features

21
Features After Selection

53% / 47%
Class Balance (No/Yes)

Behavioral (7)

Binary features capturing protective actions like hand washing and face masks

Opinion (6)

Ordinal features (1-5) capturing views on vaccine effectiveness and disease risk

Clinical (4)

Doctor recommendation, chronic conditions, health worker and health insurance

Demographic (4)

Age group, education, income and employment status

KEY EDA FINDINGS

Uncertainty Blocks Vaccination

People who 'don't know' about side effects have the LOWEST vaccination rate (18.1%) — lower than even those who are worried

Age is Critical

65+ years: 67.4% vaccinated vs 18-34 years: 28.5% — almost 40 percentage points difference

Income Predicts Vaccination

Below poverty: 36.3% vs above $75,000: 49.7% — socioeconomic barriers are real

Protective Behavior Signal

People who wash hands (49.1%) and avoid touching face (50.7%) are more likely to vaccinate than those who don't

DATA PREPARATION

1	2	3	4	5
Drop 14 Irrelevant Features	Train Test Split (80/20)	Impute Missing Values	Scale Numeric Features	Encode Categorical Features

Why this order matters - preventing data leakage

✓ Dropping columns before splitting is safe - no statistics are calculated, just removing columns

✓ Splitting before imputing prevents test set values from influencing the median used to fill training data

✓ Fitting the preprocessor on training data only then applying to test set - this is the correct approach

✓ 30 out of 36 columns had missing values - proper imputation was critical to model performance

MODELING APPROACH

01	**Logistic Regression**	Baseline	Simple, interpretable benchmark. No tuning applied to establish true baseline performance
02	**Logistic Regression + Ridge**	Tuned	L2 regularization shrinks all coefficients to reduce overfitting. C parameter tuned with GridSearchCV
03	**Logistic Regression + Lasso**	Tuned	L1 regularization shrinks some coefficients to zero performing automatic feature selection
04	**Decision Tree**	Nonparametric	Captures non-linear relationships. max_depth tuned to prevent overfitting
05	**Random Forest**	Ensemble	Combines 100-200 trees using bagging. Most robust model — our best performer

MODEL RESULTS

Model	ROC AUC	Result
Logistic Regression Baseline	**0.8506**	Good
Logistic Regression + Ridge	**0.8506**	Good
Logistic Regression + Lasso	**0.8506**	Good
Decision Tree	**0.8356**	Fair
Random Forest ★ BEST	**0.8518**	Best

BEST MODEL

Random Forest

ROC AUC

0.8518

Significantly better than random (0.5)

EVALUATING OUR MODEL PROPERLY

Beyond ROC AUC — Four Additional Evaluation Factors

Factor	Random Forest	Logistic Regression
ROC AUC	**0.8518**	0.8506
Runtime	**Slow**	**Fast**
Explainability	**Low**	**High**
Parsimony	**Low**	**High**
Ease of Deployment	**Complex**	**Simple**

✅ No Data Leakage

All transformers fitted on training data only — imputation, scaling and encoding applied to test set using training statistics exclusively

Key Insight

ROC AUC difference between Random Forest and Logistic Regression is only

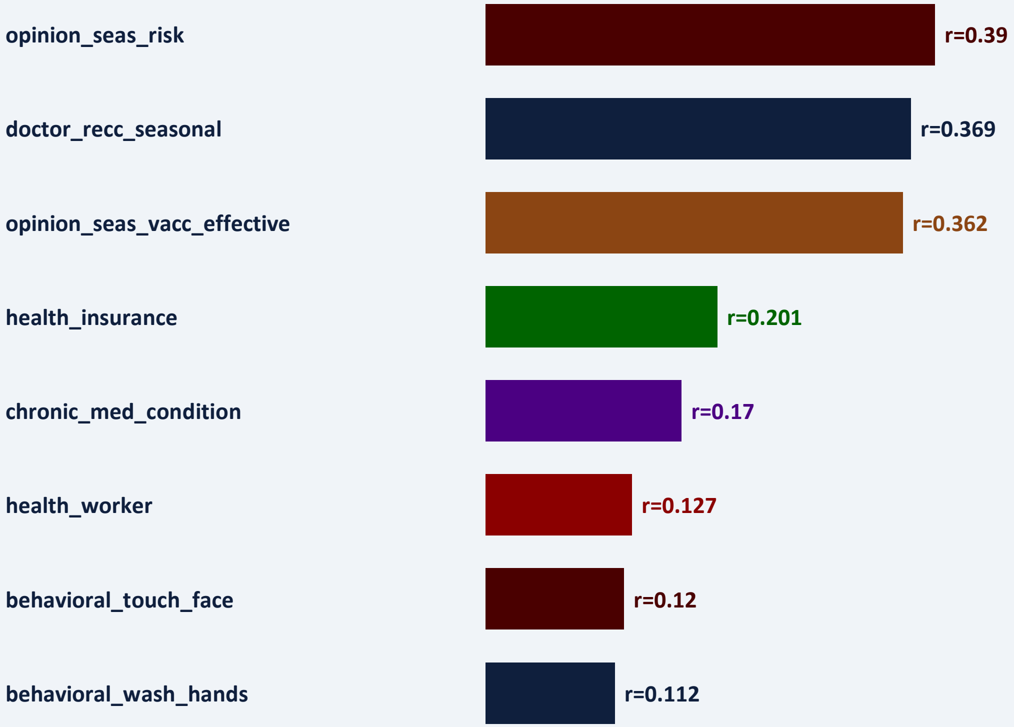
0.0012

Practically negligible

In a resource constrained Kenyan clinical setting, Logistic Regression + Lasso is a strong alternative

WHAT DRIVES VACCINATION?

Top Predictors from Random Forest Feature Importance



RECOMMENDATIONS : COUNTY DIRECTOR OF HEALTH

Target Low Risk Perception

opinion_seas_risk is the strongest predictor. Communities that perceive low flu risk are significantly less likely to vaccinate. Deploy risk education campaigns before each flu season using community health workers.

Focus on Younger Populations

18-34 years: only 28.5% vaccinated vs 65+ years: 67.4%. Partner with universities and workplaces. Consider mobile vaccination units at youth frequented locations.

Remove Financial Barriers

Below poverty: 36.3% vs above poverty: 49.7%. Provide free seasonal flu vaccination at all public health facilities. Target unemployed populations (lowest at 30.2%).

Counter Misinformation

Belief in effectiveness (r=0.362) is a top predictor. Use trusted community leaders as vaccine champions. Focus on informing not just reassuring communities.

RECOMMENDATIONS : COUNTY REFERRAL HOSPITAL CEOs

Mandate Doctor Recommendations

doctor_recc_seasonal (r=0.369) is the second strongest predictor. Implement a clinical protocol requiring all clinicians to proactively recommend seasonal flu vaccination during every consultation.

Target Uninsured Patients

health_insurance (r=0.201) is a strong predictor. Offer free seasonal flu vaccination regardless of insurance status. Flag uninsured patients for proactive outreach before each flu season.

Prioritize Chronic Condition Patients

chronic_med_condition (r=0.170). Ensure all chronic condition patients are contacted before each flu season. Integrate vaccination into chronic disease management protocols.

Address Side Effect Uncertainty

Patients who say 'Don't Know' about side effects have 18.1% vaccination — the lowest. Train clinicians to proactively address uncertainty not just worry. Provide clear written information.

MODEL LIMITATIONS

Data Limitations

- US 2009 data - may not perfectly represent the Kenyan population
- Self-reported survey data may contain response bias
- Dataset skews older, white and educated
- health_insurance has 45.96% missing values

Model Limitations

- Historical 2009 data - behavior may have changed
- Random Forest is less interpretable than Logistic Regression
- Model fairness across subgroups was not tested
- Predicts likelihood - cannot determine causality

Scope Limitations

- Does not account for vaccine availability and cost in Kenya
- Geographic variables were US-specific and had to be dropped
- A Kenya-specific dataset would improve local applicability
- Clinical validation in Kenyan hospitals is needed

IMPLEMENTATION MATRIX

Immediate Actions

→ Share findings with CDoH and Hospital CEO

→ Pilot community outreach targeting low vaccination groups

→ Implement doctor recommendation clinical protocol

→ Develop patient educational materials on vaccine effectiveness

Future Research

→ Collect Kenya-specific vaccination survey data

→ Test model fairness across demographic subgroups

→ Add county, urban/rural and distance to facility features

→ Investigate causal relationships between predictors and vaccination

Model Improvement

→ Collect more recent data beyond 2009

→ Explore Gradient Boosting and Stacking ensembles

→ Build web app for health worker use at point of care

→ Retrain annually as new vaccination data becomes available

KEY TAKEAWAY

Opinion, access and doctor recommendations drive vaccination.

Addressing these three factors through targeted community outreach and clinical protocols can significantly increase seasonal flu vaccination rates in Kenya.

THANK YOU!